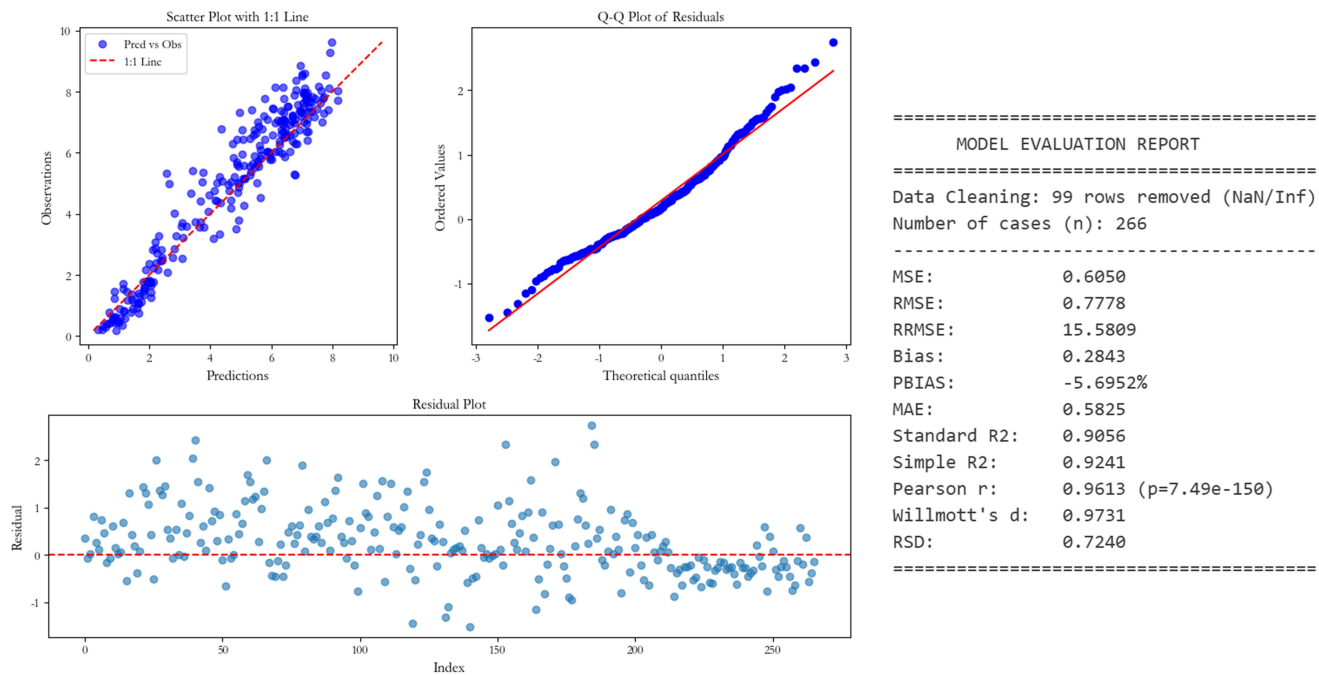




met_gfit: A Python Toolkit for Comprehensive Model evaluation and Diagnostic Plotting

A streamlined Python toolkit for comprehensive model evaluation. met-gfit automates the calculation of 10 key statistical metrics and generates professional diagnostic plots (1:1 scatter, Residual, and Q-Q plots) to validate and visualize model accuracy with ease.

Toolkit installation

```
pip install "git+https://github.com/RuiGao9/met_gfit.git"
```

Toolkit inputs

- At minimum, you need to provide two vectors:
 - `true`: observed (actual) values
 - `pred`: predicted values
- Optional inputs allow you to control the number of decimal places for statistics and whether to generate plots.

Toolkit usage example

Please refer to `test.ipynb` under the `test` folder. Or refer to below:

```
import numpy as np
from met_gfit import gfit

obs = [1, 2, np.nan,
```

```

    4, 5, 5.5,
    4.8, 6.1, 5.0,
    3.3, 1.7, 2.5]
sim = [1.1, 1.9, 6,
       3.8, 5.1, 5.4,
       np.nan, 6.0, 5.2,
       np.nan, 1.5, 2.7]

val = rmse(obs, sim)
all_stats = gfit(obs, sim, plots='No', show_results='No')

# With more data points and some noise
np.random.seed(42)
# Create a larger dataset with 1000 samples
obs = np.linspace(0, 100, 1000)
noise = np.random.normal(0, 5, 1000)
bias = 1.3
sim = obs + noise + bias
# Add some NaN and Inf values randomly
nan_indices = np.random.choice(range(1000), 10, replace=False)
obs[nan_indices] = np.nan
# Add some Inf values randomly
inf_indices = np.random.choice(range(1000), 5, replace=False)
sim[inf_indices] = np.inf

# Showing the results
results = gfit(obs, sim, num_decimal=4)

```

Comprehensive Evaluation

The core function `gfit()` performs a full-scale diagnostic of your model, providing:

- **11 Statistical Metrics:** Returns a dictionary/report containing R^2 , $RMSE$, MAE , $Bias$, $PBIAS$, r , d -index, and more.
- **Visual Diagnostics:** Optionally generates a professional three-panel figure including:
 - **1:1 Scatter Plot:** Visualizes accuracy and potential scaling issues.
 - **Residual Plot:** Checks for non-linearity and heteroscedasticity.
 - **Q-Q Plot:** Evaluates the normality of the error distribution.

File Structure

The repository is organized following the standard Python `src` layout for better compatibility and installation:

- `src/met_gfit/`:
 - `metrics.py`: The engine containing all mathematical formulas and plotting logic.
 - `__init__.py`: The gateway that allows for clean imports (e.g., `from met_gfit import gfit`).
- `pyproject.toml`: Project metadata and dependency specifications.
- `README.md`: This comprehensive documentation and user guide.
- `LICENSE`: MIT License for open-source use.

Considered goodness-of-fit statistics

10 statistics are included in this function "gfit.py". They are:

- **MSE**: Mean Squared Error. Square of the unit of the output variable.
Measures the **average squared difference** between observed and predicted values. Lower values indicate a better fit.
- **RMSE**: Root Mean Square Error. The unit is the same as output variable.
The square root of MSE; shows the **average error** in the same units as the data.
- **Bias**: Bias. The unit is the same as output variable.
The average difference between predicted and observed values. Indicates if predictions are systematically too high or too low.
- **PBIAS**: Percentage Bias. The unit is %.
Measures the average tendency of the prediction to be larger or smaller than the corresponding observed objective.
 - $|PBIAS| < 10\%$: Very Good
 - $10\% \leq |PBIAS| < 15\%$: Good
 - $15\% \leq |PBIAS| < 25\%$: Satisfactory
 - $|PBIAS| \geq 25\%$: Unsatisfactory
- **r**: Correlation coefficient. Unitless.
Measures the strength and direction of the linear relationship between observed and predicted values (ranges from -1 to 1).
- **p-value**: p-value. Unitless.
 - It tells how likely it is that the result happened by random chance.
 - A **small p-value** (usually less than 0.05) means the result is probably not just random, so there is likely a real effect of the relationship.
 - A **large p-value** means the result could easily happen by chance, so there may not be a real effect.
- **d**: Willmoot's index of agreement. Unitless.
Measures the degree of model prediction error (ranges from 0 to 1, with 1 being perfect agreement).
- **R2**: Coefficient of determination. Unitless.
Shows the proportion of variance in observed values explained by the predictions (ranges from 0 to 1).
- **MAE**: Mean Absolute Error. The unit is the same as the output variable.
The average of the absolute differences between observed and predicted values.
- **RRMSE**: Relative Root Mean Square Error. %.
RMSE expressed as a percentage of the mean observed value.
- **RSD**: The standard deviation of the residual. The unit is the same as the output variable.
Indicates how spread out the residuals (errors)

Equations

met_gfit provides a comprehensive assessment of model performance, covering accuracy, precision, and agreement with observations. The following statistics are implemented:

Mean Square Error (*MSE*)

Measures the average of the squares of the errors, giving more weight to larger deviations (outliers).

$$MSE = \frac{1}{n} \sum_{i=1}^n (o_i - p_i)^2$$

Root Mean Square Error (*RMSE*)

Indicates how much the model's predictions deviate from the observations on average.

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (o_i - \hat{p}_i)^2}{n}}$$

Mean Bias (*Bias*)

Measures the average tendency of the model to over- or under-estimate the observed values. A positive value indicates overestimation.

$$Bias = \frac{1}{n} \sum_{i=1}^n (p_i - o_i)$$

Percentage Bias (*PBIAS*)

Measures the average tendency of the prediction to be larger or smaller than their observed values.

$$PBIAS = \frac{\sum_{i=1}^n (p_i - o_i)}{\sum_{i=1}^n o_i} \times 100$$

Pearson Correlation Coefficient (*r*)

Indicates the strength and direction of the linear relationship between predicted and observed values. A correlation close to 1 or -1 indicates that the model captures the variability of observations well.

$$r = \frac{\sum (o_i - \bar{o})(p_i - \bar{p})}{\sqrt{\sum (o_i - \bar{o})^2 \cdot \sum (p_i - \bar{p})^2}} = \frac{cov(o, p)}{\sigma_o \sigma_p}$$

Notes: In EXCEL, the R^2 value (via the **RSQ** function) represents the squared Pearson correlation coefficient (r^2). This metric quantifies the degree of linear correlation between observations (o) and predictions (p). In contrast, the standard **Coefficient of Determination** evaluates the predictive accuracy by accounting for systematic biases.

The Excel workbook **R_Square_Calculation.xlsx** provides a comparative example demonstrating the discrepancy between Excel's default R^2 calculation (**Pearson's r^2**) and the standard **Coefficient of Determination** ($1 - SS_{res}/SS_{tot}$).

$$R^2 = r^2 = \left[\frac{\sum (o_i - \bar{o})(p_i - \bar{p})}{\sqrt{\sum (o_i - \bar{o})^2 \cdot \sum (p_i - \bar{p})^2}} \right]^2$$

Willmott's Index of Agreement (*d*)

Considers both the mean absolute error and the variability of the observations, showing insight into how well the model replicates the variability and pattern of the observations (Willmott, 1982).

$$d = 1 - \frac{\sum_{i=1}^n (o_i - p_i)^2}{\sum_{i=1}^n (|p_i - \bar{o}| + |o_i - \bar{o}|)^2}$$

Coefficient of Determination (R^2)

Quantifies the proportion of the variance in the observations that is predictable from the model, providing a rigorous measure of model fit.

$$R^2 = 1 - \frac{\sum_{i=1}^n (o_i - \hat{p}_i)^2}{\sum_{i=1}^n (o_i - \bar{o})^2} = 1 - \frac{SS_{res}}{SS_{tot}}$$

Mean Absolute Error (MAE)

Represents the average of the absolute differences between observations and predictions, providing a linear score where all individual differences are weighted equally.

$$MAE = \frac{1}{n} \sum_{i=1}^n |o_i - p_i|$$

Relative Root Mean Square Error ($RRMSE$)

Normalizes the $RMSE$ by the mean of the observations, expressed as a percentage, which allows for comparison between different datasets or variables.

$$RRMSE = \frac{RMSE}{\bar{o}} \times 100\%$$

Residual Standard Deviation (RSD)

Quantifies the spread of the residuals (the differences between model and observations) around their mean.

$$RSD = \sqrt{\frac{\sum_{i=1}^n (p_i - o_i - \frac{1}{n} \sum_{i=1}^n (p_i - o_i))^2}{n - 1}}$$

where o represents observations, o_i is the observed value for the i th observations, $\bar{o}$ is the mean of observations, p represents predictions, p_i is the estimated value for the i th estimations, $\hat{p}_i$ is the estimated value corresponding to the i th observations, n is the number of paired observations.

Reference

- Willmott, C. J. (1982). Some comments on the evaluation of model performance. Bulletin of the American Meteorological Society, 63(11), 1309-1313. [https://doi.org/10.1175/1520-0477\(1982\)063<1309:SCOTEO>2.0.CO;2](https://doi.org/10.1175/1520-0477(1982)063<1309:SCOTEO>2.0.CO;2)
- Nash, J. E., & Sutcliffe, J. V. (1970). River flow forecasting through conceptual models part I — A discussion of principles. Journal of Hydrology, 10(3), 282-290. (For NSE/R^2 implementation).

Software Citation

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@misc{gao2025met_gfit,  
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}
```

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